

Course Name: Computer Organization

Course Outcome

- CO1- Understand the basics of digital computer system.
CO2- Demonstrate the principle of arithmetic operations on unsigned, signed integers and floating point numbers.
CO3- Understand the concepts of combinational and sequential circuits and their applications.
CO4- Understand the CPU architecture and organization.
CO5- Explain the basic concepts of pipelining.
CO6- Design the steps for the execution of the complete instruction for hardwired and micro-programmed control unit.
CO7- Explain the function of memory hierarchy.
CO8- Determine the interface of CPU with input/output devices and their modes of transfer.

Printed Pages: 03

University Roll No.

End Term Examination, Odd Semester 2023-24 **B. Tech. (CS/AIML/CCV/CSF/DA/IIOT), II Year, III Semester** **BCSC 1005 : Computer Organization**

Time: 3 Hours

Maximum Marks: 50

Section – A

Attempt All Questions

4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider two binary numbers $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ are stored in two registers A & B based on Parallel-In Parallel-Out mode respectively. Design a combinational logic to subtract two numbers.	4	3	U	C
2	Show how a computer processor evaluates the following arithmetic expression by using reverse polish notation? $((54-20)*((100-3*10)/7)/2)+15$	4	4	A	P
3	Represent $(175.51)_2$ in single and double precision floating point number.	4	2	A	P
4	Consider a cache consisting of 256 blocks of 16 words each for a total of 4096 words. Also, assume that the main memory is addressable by a 16 bits address and accommodate 12 bits data. How many bits are there in each of Tag, Index, Block, and Word field for 2-way set associative technique? Include a valid bit. What is the size of the cache memory?	4	6	U	P
5	Consider a pipeline having 4 phases with duration 60, 50, 90 and 80 ns. Given latch delay is 10 ns. Calculate-	4	5	U	P

1. Pipeline cycle time				
2. Non-pipeline execution time				
3. Speed up ratio				
4. Pipeline time for 1000 tasks				
5. Sequential time for 1000 tasks				
6. Throughput				
Or				
In a certain scientific computation, it is necessary to perform the following arithmetic expression.				
$A_i * \left(\frac{B_i}{C_i}\right) + D_i$				
(a) Specify the pipeline configuration to carry out the above task.				
(b) List the content of all registers in a space-time diagram for 5 tasks.				

Section – B

Attempt All Questions

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Define (a) Locality of reference (b) Write-back (c) Page fault	3	5	R	F
7	A computer uses RAM chips of 1024x1 capacity. (a) How many chips are needed, and how should their address lines be connected to provide a memory capacity of 1024 bytes? (b) How many chips are needed to provide a memory capacity of 16K bytes?	3	7	U	C
8	Define (a) Virtual memory (b) Instruction cycle (c) Interrupt	3	6	R	F
9	Explain different methods of asynchronous data transfer.	3	8	R	F
10	What sequence of micro-operations is performed by a computer system to execute the STA instruction for which two references to the memory are required to compute the memory address?	3	7	U	C

Section – C

Attempt All Questions

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	A virtual memory system has an address space of 8K words, a memory space of 4K words, and page & block sizes are of 1K words. The following page reference changes occur during a given time interval. 3 0 2 1 2 1 3 4 0 1 0 3 2 5 6 0. Determine the four pages that are resident in main memory after each page reference change if the replacement algorithm used is (a) FIFO (b) LRU	5	7	U	P
12	Explain a DMA controller and its mode of data transfer with proper illustration.	5	8	R	F
13	A computer employs RAM chips of 256x8 and ROM chips of 1024x8. The computer system needs 2K bytes of RAM, 4K bytes of ROM, and four interface units each with four registers. A memory-mapped I/O configuration used. The two highest-order bits of the address bus are assigned 00 for RAM, 01 for ROM, and 10 for interface registers. (a) How many RAM and ROM chips are needed? (b) Draw a memory-address map for the system.	5	7	U	P